

Introduction

Overview

Demographics

Market

Product

Segment Analysis

Retention

Strategy

Customer Segmentation Report

Executive Summary

- **The Problem:** A global electronics retailer experience a big drop in Revenue(49.1%) after growth, peaking at \$18.3M in 2019.
- **The Project goal:** identify distinct customer groups based on purchasing behavior to support targeted marketing and retention strategies.
- **Data:** orders from 2016 to Feb 2021 across 8 countries (online and in-store). Analysis reference date: 1st April 2021.

15,266

Registered
Customers

2,517

Products

26,326

Distinct Orders

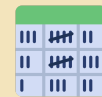
67

Stores

Key Metrics:



Recency: Quarters since last order



Frequency: Orders per quarters (relative to customer tenure)



Monetary: Total Spending (Revenue)



Active Span: time between 1st & last order in quarters

Data Processing:



MySQL: data cleaning and RFM segmentation



Power BI: building visualization and interactive reporting

Segment



At Risk
Champions
Hibernating
Loyal Customers
Must Keep
New Customers
Potential Loyal
Promising

Introduction

Overview

Demographics

Market

Product

Segment Analysis

Retention

Strategy

country

All

Product Categ...

All

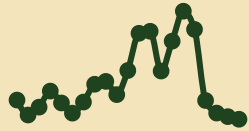
Order Type

All

Overview

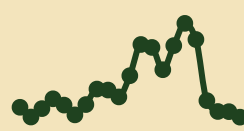
55.76M

Total Revenue



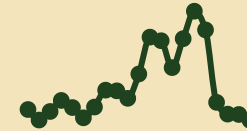
11887

Active Customers



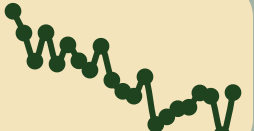
26326

Orders



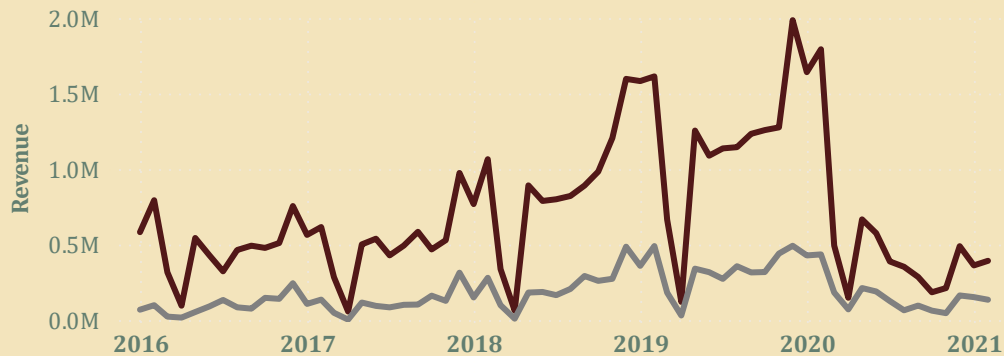
2,117.89

Average Order Value



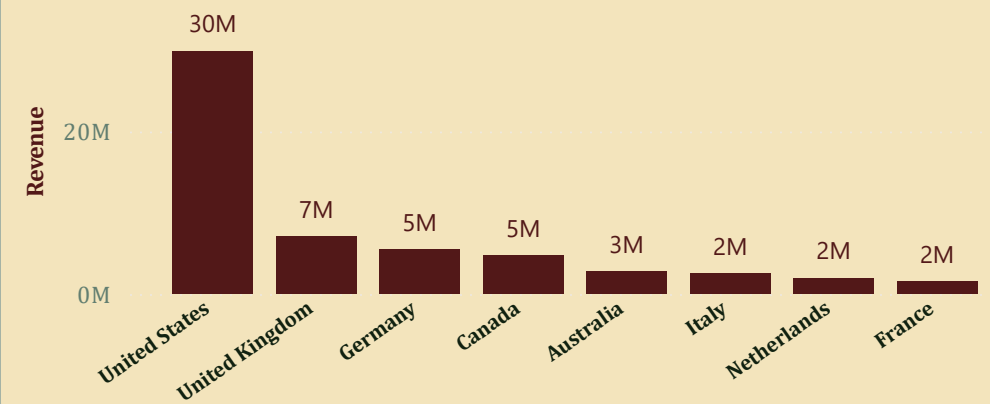
Revenue by Order Type

order_type ● In-store ● Online



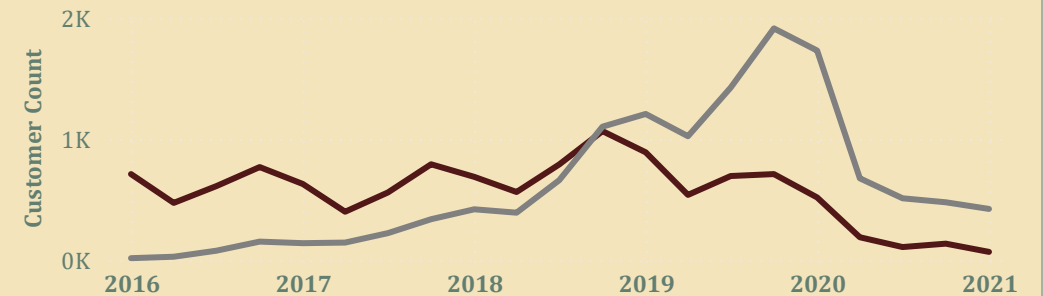
- **Revenue** peaked at \$18.3M in 2019, dropped 49.1% to \$9.3M in 2020.
- **New customer** acquisition dropped in starting 2019, but **Revenue growth** display the **same increasing pattern** as **previous years**. Only shows **drastic drop** starting 2020
- **US** contributes ~54% of total **revenue**, largest single market.
- **In-store** dominates revenue but **Online** showed more resilience during 2020.

Revenue by Country



New/Return Customer count

Type ● New ● Return



Introduction

Overview

Demographics

Market

Product

Segment Analysis

Retention

Strategy

country

All

Year

All

Order Type

All

Demographic

55.61M

Male Revenue

6029

Male Count

55.60M

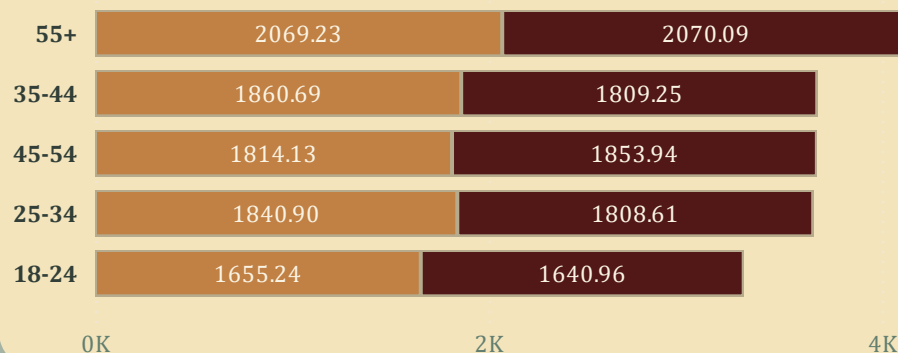
Female Revenue

5858

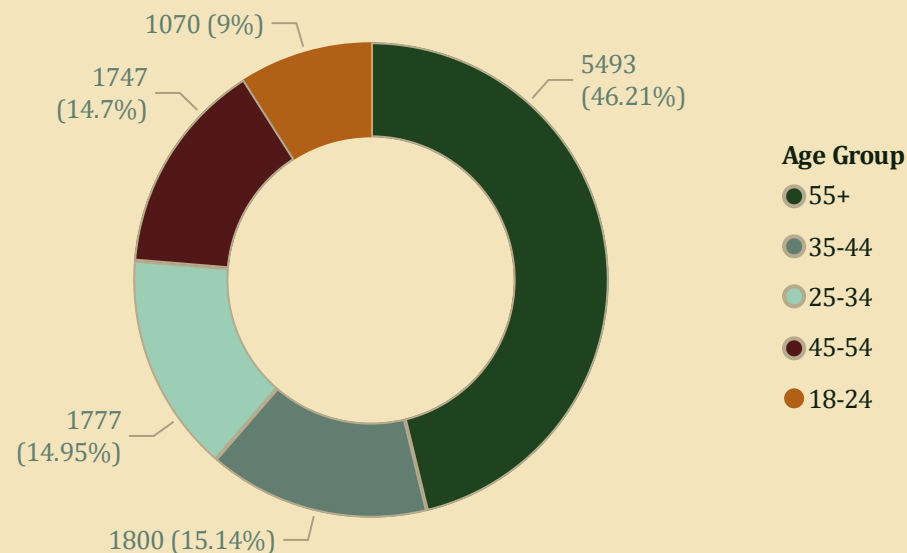
Female Count

AOV by Age Group

Age Group ● Female ● Male

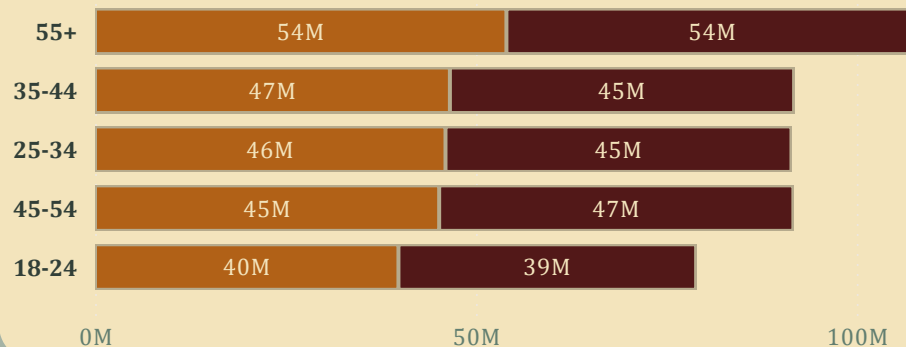


Customer count by Age Group



Revenue by Age Group

gender ● Female ● Male



- **55+** group generate ~**46%** total **Revenue** while also being the biggest group, but their **AOV** does not dominate => leading through volume.
- **Gender** split nearly equal across **Revenue** and **Count**.
- **18-24** group have **highest long-term growth**, but currently the **smallest** group.
- **AOV** showing similar spending behaviors across **Age Group**, **acquisition** is likely the cause of difference in **Revenue**.

Introduction

Overview

Demographics

Market

Product

Segment Analysis

Retention

Strategy

Country

All

Gender

All

Year

All

Order Type

All

Market Analysis

Market Comparison

All

All

11887

Customer count

11887

55.76M

Total Revenue

55.76M

26326

Order Count

26326

2,117.89

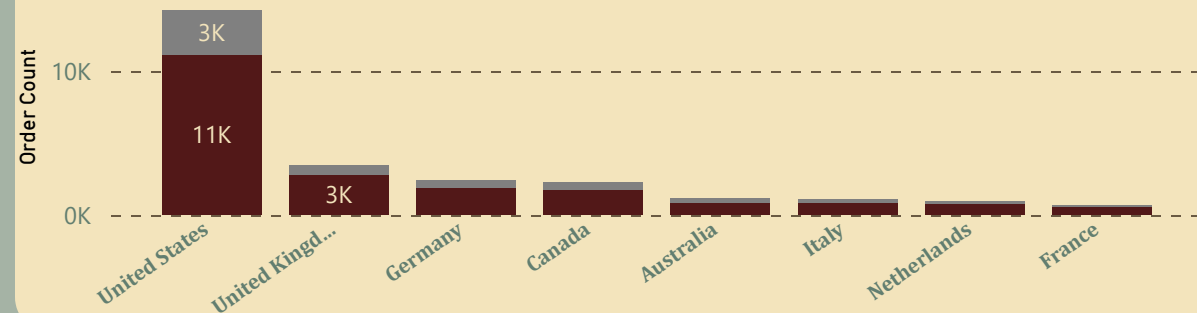
AOV

2,117.89

- **US:** highest **Order Volume** & **Return Rate (40.67%)** but **AOV** sits below average => Most **Loyal Market** lead through **volume**.
- **Aus** and **Italy:** biggest drop in **2019-2020** (~50-53%) in all aspect.
- **France:** smallest drop in both **new (-48%)** and **returning (-10%)** customers during the Pandemic.
- **Online order:** minimal across markets => **underutilized**.
- **New Customers** dominate => **Retention** is key.
- **UK** and **Germany:** perform best in **Europe** best **return**

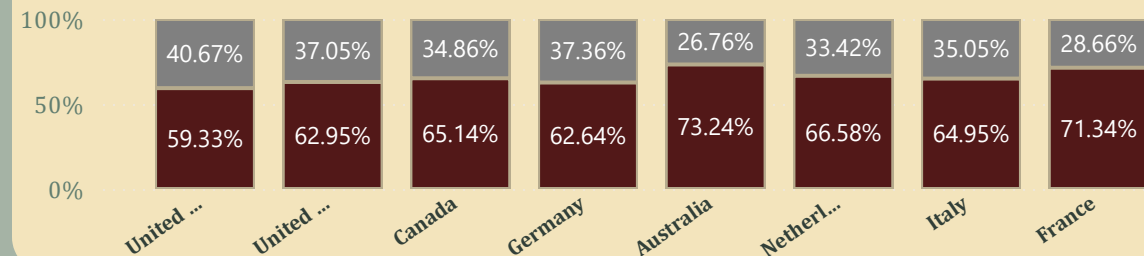
Orders Count by Country

order_type ● In-store ● Online



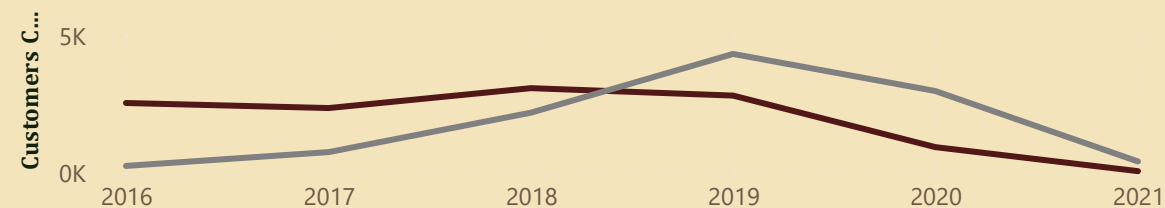
Customer Distribution by Country

Type ● New ● Return



Customer by Type

type ● New ● Return



Introduction

Overview

Demographics

Market

Product

Segment Analysis

Retention

Strategy

Year

All

New/Return

All

Order Type

All

Category

All

Product Analysis

Category Comparison

Computers

Cell phones

7459

Customer count

6253

45.42M

Total Revenue

37.35M

10990

Order Count

8442

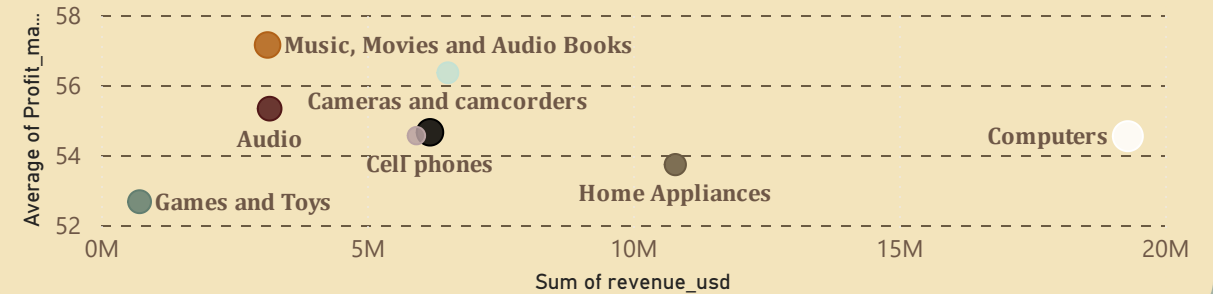
1,756.29

AOV

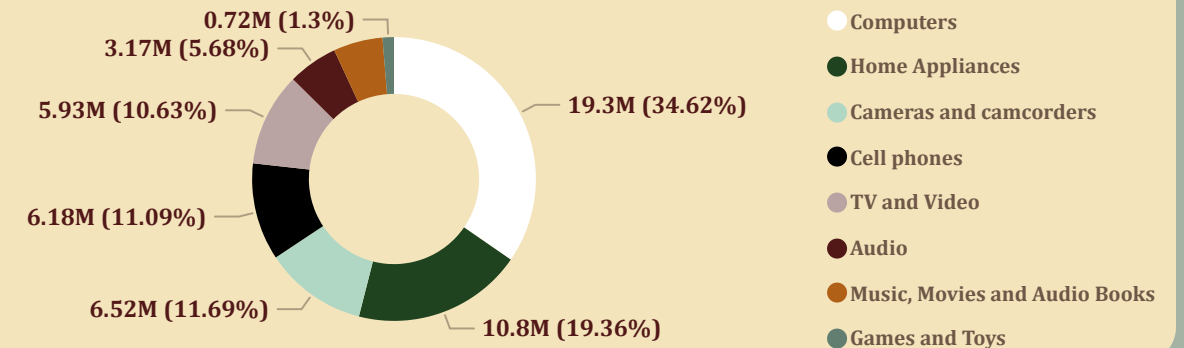
732.50

- **Computers** dominates **Revenue (\$19.3M)**, but has average **profit margin (~54%)**.
- **Music, Movies & Audio Books** and **Cameras** have the **highest margins (~57%)** but significantly lower revenue.
- **Home Appliances** generates **2nd highest Revenue (\$11M)** with the **2nd lowest Margin => low efficiency**.
- **Games and Toys** is the weakest => **lowest priority**.
- **All categories** decline in **2020** => **systemic issue**, not **category-specific**.

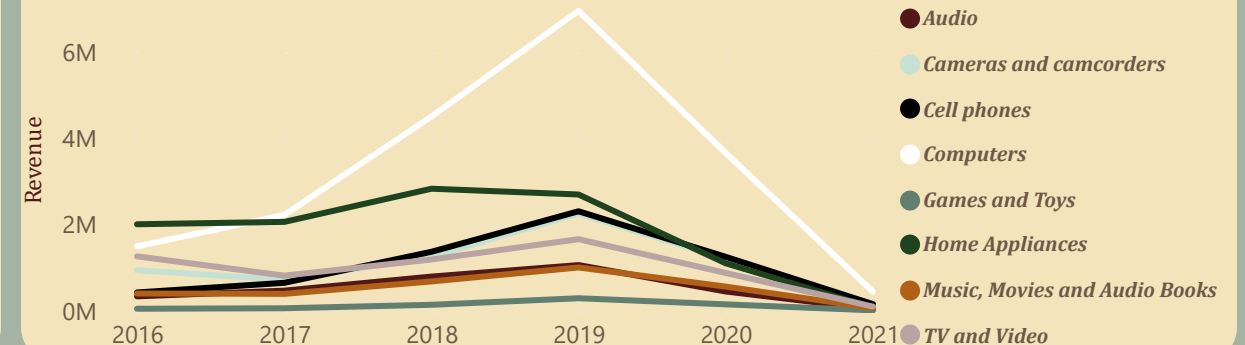
Revenue vs Average Profit Margin



Revenue by Product Category



Revenue by Category over time



Introduction

Overview

Demographics

Market

Product

Segment Analysis

Retention

Strategy

Country

All

Age Group

All

Gender

All

Segment

All

Segment Analysis

Segment Comparison

All

11887

55.76M

4,690.46

100.00

Customer
count

Total
Revenue

Average
Spending

% of total
Revenue

All

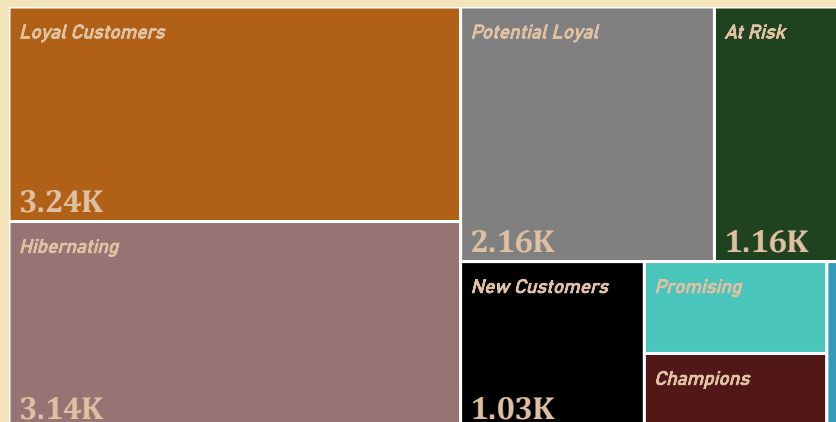
11887

55.76M

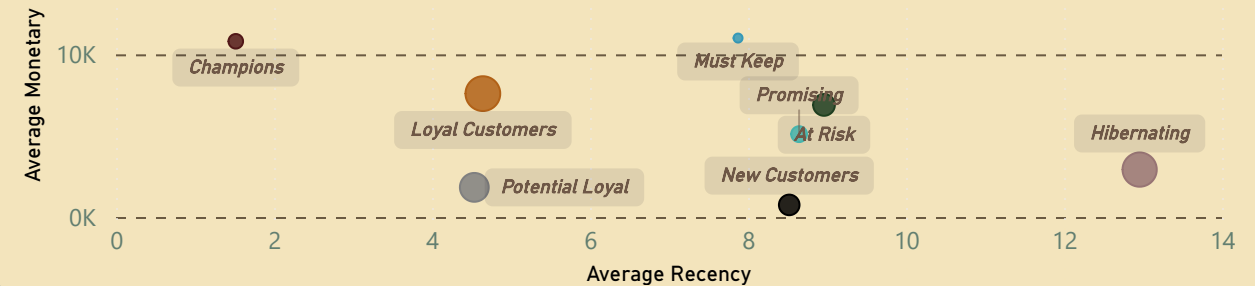
4,690.46

100.00

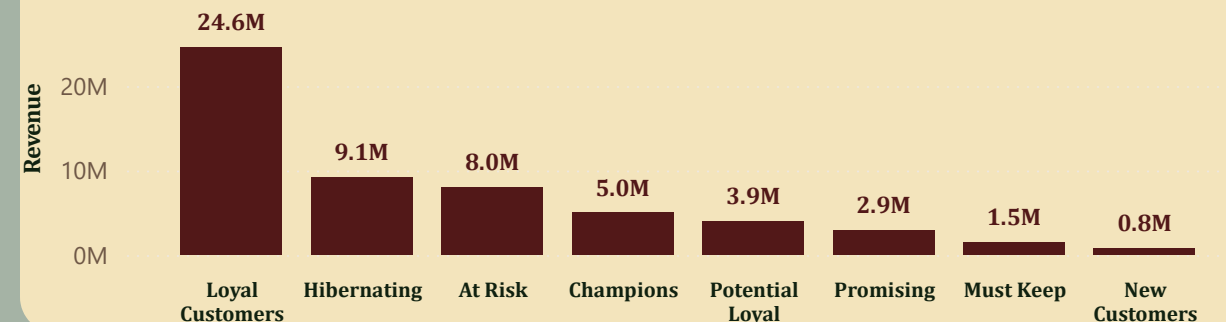
Segments Profile Matrix



Segments Profile Matrix



Revenue by Segment



- **Loyal Customers:** the 2nd largest group (3241), Highest **Revenue** (\$24.6M) with lower spending than **Champions'** =>Volume Driven.
- **At Risk** generates \$8M with only 1163 customers =>good **value** for recovery.
- **Hibernated:** 2nd biggest group, \$9.1M in **Revenue** =>High recovery value
- **Must Keep:** smallest group, low **Recency**, but highest **Average Spending** => critical to recover.
- **New & Promising Customers:** decent size but low in **Revenue** => **Conversion** is key.

Introduction

Overview

Demographics

Market

Product

Segment Analysis

Retention

Strategy

year

All

Quarter

All

Segment

All

Quarter After

All

Retention Rate

7.75

Avg 1-Quarter Retention R...

7029

Post 1st-quarter-Returner

20.55

Max of retention_rate

566.05

Avg qtrly new Customer

New Customer Count by Quarter

year	1.00	2.00	3.00	4.00	5.00	6.00	7.00	8.00	9.00	10.00
⊕ 2016	125	143	147	153	176	191	232	250	298	351
⊕ 2017	148	180	205	250	283	290	351	393	348	283
⊕ 2018	355	376	435	457	485	390	323	208	137	102
⊕ 2019	386	354	342	213	124	80	56	25		
⊕ 2020	30	22	22	15						
Total	1044	1075	1151	1088	1068	951	962	876	783	736

Retention Rate by Quarter

year	1.00	2.00	3.00	4.00	5.00	6.00	7.00	8.00
⊕ 2016	5.08	5.88	5.50	5.83	6.65	7.38	9.00	9.36
⊕ 2017	6.17	7.74	8.43	9.88	11.88	12.54	14.74	16.15
⊕ 2018	10.88	12.36	14.11	14.27	15.31	13.75	11.61	7.18
⊕ 2019	13.72	12.61	11.79	6.96	4.26	3.63	3.70	2.81
⊕ 2020	2.90	2.86	3.65	2.90				
Average	7.75	8.58	9.26	8.86	9.52	9.71	10.63	10.27

- **Average retention** of 7.75% after 1 quarter => ~92% of customers do not return the next quarter after their 1st purchase.
- **2019** has the best **1-quarter retention** (13.72%), while it collapse in **2020** (2.90%)) => **Pandemic** impact.
- **At Risk**: 1Q rate is low (2.82%) but rises to 7.63-11.72% after 2-5 quarters => **optimal reach out window**.
- **Must Keep & Champion**: no 1Q rate => high value **segments** have a longer **repurchase cycle**.
- **Hibernating**: Higher rate than **Loyal & Potential Customers** => **recently Hibernated** are more responsive than expected, potential for reactivated.
- **Loyal Customer**: steep decline in **Retention Rate** in 2020 => prioritized for **re-engagement campaigns**.

Introduction

Overview

Demographics

Market

Product

Segment Analysis

Retention

Strategy

Recommendation

PROTECTING

Champions and Loyal Customers:

- Exclusive offer, early access/release.
- Set up **Recency** warning, personal reach-out after about every **2-3 quarters**.
- Target **55+** group in the segments, for both **Genders**.

US Market: Prioritize for protecting. target all segments here.

Computer and Home appliances: Most sales with decent profit, promote sales/offers for new release and upgrades.

WIN BACK

At Risk: highly reactive within **2-4 quarter** (~7-11% **Retention Rate**) => best time to win back.

Must Keep: Highest **ROI recovery** target (**highest average spending**), small group so can leverage personalized outreach, target **Female-specific** message for more **Revenue**.

Lapsing spend more, less often, push for **Category** relevance.

Australia and Italy: steepest **customer losses** in 2020 — **highest priority** markets for **win back** campaigns.

REACTIVATING

Hibernated:

- Can target immediate after **1-4 quarters**.
- Lead with **New drops** to engage with them (High **Revenue**, just long **inactivity**).

Online: better chance to reach **dormant Customers**.

Music/Movies and Cameras: Highest **Profit Margin** (~56-57%) — good fit for **reactivation campaign** with.

DEVELOPING

New Customer & Promising

- ~97% of **New Customers** don't come back after **1st quarter** => leverage the **1st 90 days**, aggressive follow up offers recommended.
- Revenue is not high => prioritized conversion.

Potential Loyal: quite **stable** but not committed => **offers** or **loyalty programs** invites to convert.

Online: utilize more to acquire and keep more **New Customers**, especially for younger **Demographics**.

Australia: Highest proportion of **New Customers** across **all years** => Best market for **conversion campaign**.